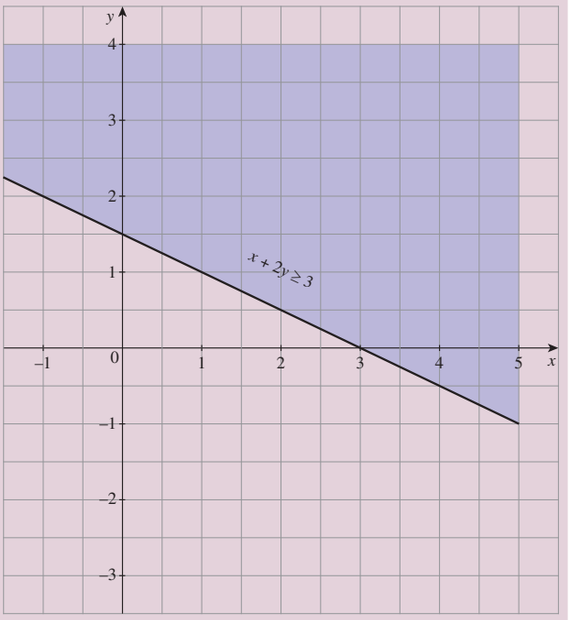
Example 2.29 Shade the region given by the linear inequality $x + 2y > 3$.

Proof. The line $x + 2y = 3$ divides the cartesian plane into two half planes. To find the half plane represented by $x + 2y > 3$ substitute a point in one of the half planes in the inequality and check whether it is satisfied. Let us substitute $(0, 0)$ in the inequality. We get $0 > 3$ which is false. Hence, the region given by $x + 2y > 3$ is the half plane which does not contain the origin.



Example 2.30 Solve the linear inequalities and exhibit the solution set graphically:

$$x + y \geq 3,\ 2x - y \leq 5,\ -x + 2y \leq 3.$$

Solution:
Observe that a straight line can be drawn if we identify any two points on it. For example, $(3, 0)$ and $(0, 3)$ can be easily identified as two points on the straight line $x + y = 3$.
Draw the three straight lines $x + y = 3, 2x - y = 5$ and $-x + 2y = 3$.
Now $(0, 0)$ does not satisfy $x + y \geq 3$. Thus, the half plane bounded by $x + y = 3$, not containing the origin, is the solution set of $x + y \geq 3$.

Similarly, the half-plane bounded by $2x - y \leq 5$ containing the origin represents the solution set of the $2x - y \leq 5$.

The region represented by $-x + 2y \leq 3$ is the half space bounded by the straight line $-x + 2y = 3$ that contains the origin.

The region common to the above three half planes represents the solution set of the given linear inequalities.

